

Experimental and theoretical investigations of affect in language and cognition

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Day 5: Emotions and Decision-making

<https://mcmoyer11.github.io/pages/essli-2026.html>

ROOM: B-286

Lecturers:

Michelle Stanković

Morgan Moyer



Course Outline

~~Day 1: intro / baby r~~

~~Day 2: psych / corpus~~

~~Day 3: psycho-ling~~

~~Day 4: sem-prag~~

Day 5: decision making

Brain areas implicated in (emotional) decision-making?
How does emotion affect downstream judgement?

How emotions affect decisions

Zajonc (1980) forced us to evaluate the question: **if I have an affective reaction, what does my cognitive system do with it?**

**EMOTION
VS REASON**

**EMOTION
AND REASON**



Affective and deliberative processes **interact** to construct value and guide action

What kinds of decisions?

- 7×8 ? *No*
- **Risk and uncertainty** (Loewenstein et al., 2001)
- **Preference and valuation** (Lerner et al., 2004)
- **Moral judgements and decisions** (Greene et al., 2001; Haidt, 2001)
- **Social judgements** (e.g. trust, blame; Dunn & Schweitzer, 2005)
- **Complex, novel or ambiguous judgements** (Forgas, 1995)

The Somatic Marker Hypothesis

- Emotion can help decisions by carrying information from **previous consequences** into the current choice:
- This information is associated with affective responses: gut reactions about it being good/bad: they are **somatic markers**
- When we face a similar choice again, those responses can be **reactivated before we decide**.
 - Gut reaction: NOPE or YAY
- Somatic markers bias attention toward or away from particular options, reducing the number of possibilities that need explicit deliberation. They are **affective cues**
 - *If you disrupt the systems that generate or use these markers then you will make poorer decisions under uncertainty.*
- See: Damasio 1994, 1996

Where is it actually going wrong?

- BUT: Damasio's strong somatic marker hypothesis is controversial and underspecified (see: Linquist & Bartol, 2013).

At which point of the decision process do we need a somatic marker?

- **Decision-point recognition** → realising that the current situation requires a decision
- **Option generation** → generating possible courses of action
- **Deliberation** → identifying and considering consequences, including deciding when to *stop* deliberating
- **Evaluation** → assigning values to options and ranking them
- **Execution** → translating the resulting decision into action
- If VMF damage destroys the emotional mechanism required for generating, deliberating over and evaluating options, why was Elliot good at laboratory tasks explicitly testing those processes?

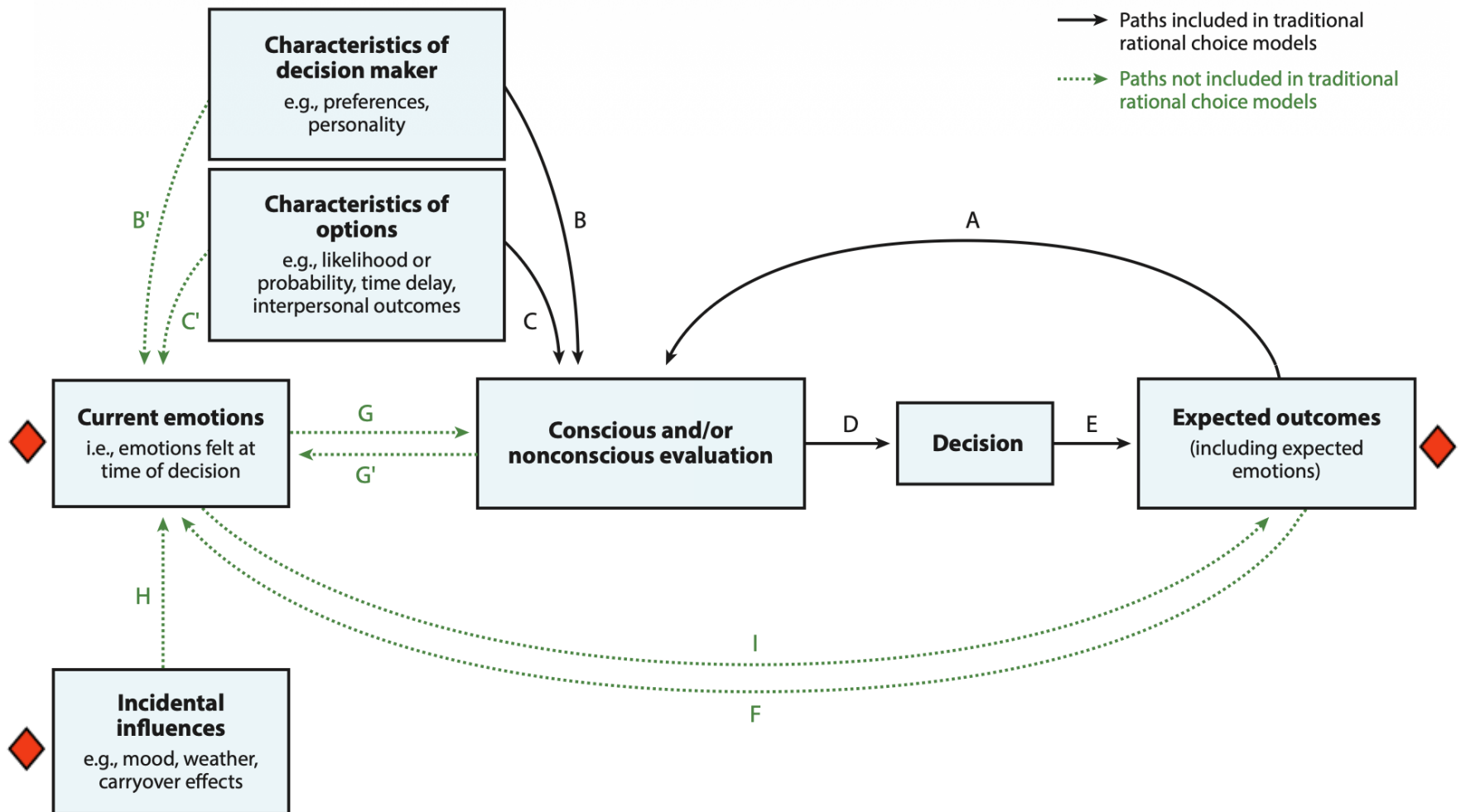


Figure 2

Toward a general model of affective influences on decision making: the emotion-imbued choice model.

Model from Lerner et al. (2015)

The case of “Elliot” / EVR

- Successful member of society by all accounts until he suffered bilateral damage to orbital and lower medial frontal cortex (OFC/vmPFC), R > L
- **Some functioning became impaired:**
 - Getting ready for work took at least 2 hours
 - Excessive time spent on minor decisions like deciding where to eat
 - Made bad investments against advice, went bankrupt
 - Couldn't hold down a job
 - Divorced after 17 years, second marriage also failed
- **Some functioning remained excellent:**
 - Superior intelligence and memory
 - Could reason appropriately about hypothetical problems
 - Retained knowledge of social rules and appropriate behaviour
- Excessively deliberative, but unable to turn reasoning into effective **goal-directed behaviour**.
- Can solve clearly specified problems, but unable to deal with competing goals – **this is where emotion helps**
- **Severe real-world decision impairment can coexist with preserved IQ, explicit reasoning and social knowledge**
- See: Eslinger & Damasio, 1985

Table 1. Neuropsychological test scores of EVR*

Intellect

WAIS-R Verbal IQ	129	(97th; superior)
Performance IQ	135	(99th; very superior)
Shipley-Hartford Vocabulary Score	37/40	(superior)
Abstractions Score	40/40	(superior)

Memory

Wechsler Memory Scale†	
Quotient	143
Digit Span (ACSS)	14.0
Logical Memory (ACSS)	18.1
Paired Associates (ACSS)	16.3

Rey Auditory Verbal Learning (max: 15)

Recall Trial 1	10
2	8
3	12
4	14
5	14
Recognition	14
Delayed Recall	11

Benton Visual Retention Test (Adm. A; Form C)

Correct	9
Errors	1

Rey-Osterrieth Complex Figure

Copy	36/36
Delayed Recall	32/36

* Percentiles and interpretation in parentheses.

† Administered without visual reproductions subtest.

If you told Elliot about a problem, he could often tell you what someone should do about it

But if you put Elliot in the equivalent real-world situation, he might make a disastrous choice.

Reasoning ability is not sufficient for adaptive real-world choice.

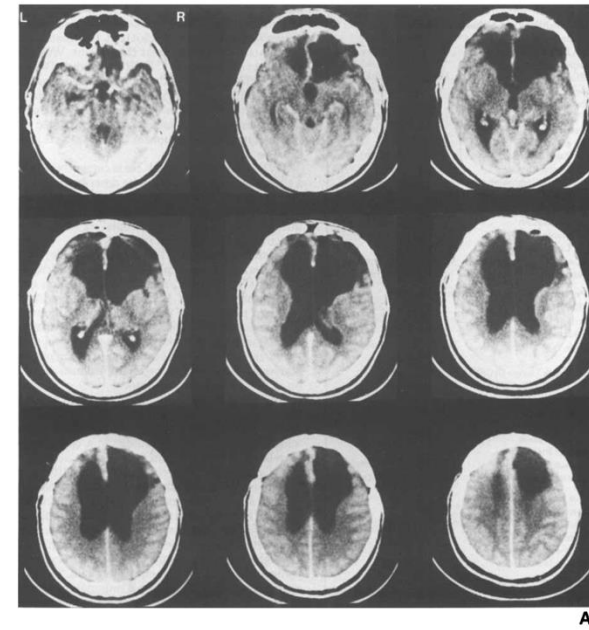
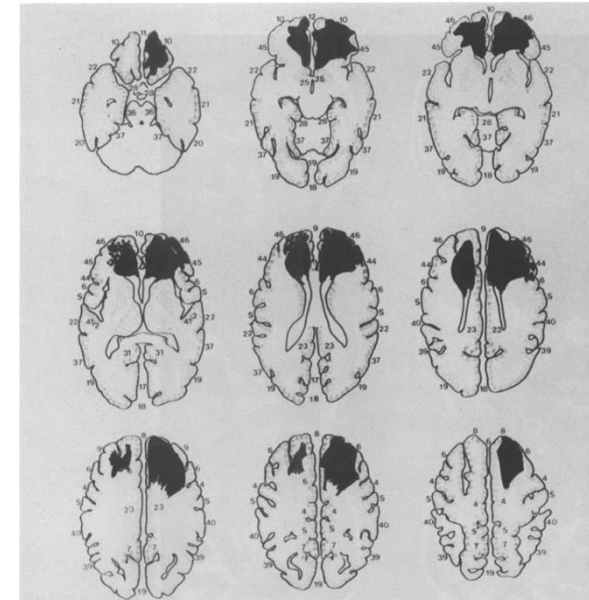


Figure 1. CT images and templates of patient EVR. Each cut was 8 mm. There was a low-density area in both frontal lobes, which merged with the image of the ventricular system, corresponding to the surgical resection. The resection involved (1) all of the orbital cortex on the right and part of the orbital cortex on the left; this lesion included both frontal poles and corresponded on the right to Brodmann's cytoarchitectonic fields 11, 12, and 25; on the left, field 25 is intact and fields 11 and 12 were only partially damaged; white matter subjacent to these orbital



fields was missing; (2) the right mesial cortex, encompassing cytoarchitectonic fields 32, 10, 8, 9, 24, and the most anterior portion of area 6; on the left, the mesial lesion involved areas 8 and 9; and (3) part of the right dorsolateral cortex (prefrontal fields 8, 9, and 46), and white matter subjacent to the premotor region. The equivalent area on the left was preserved. The basal forebrain region was preserved, as were both motor and somatosensory cortices. The temporal, parietal, and occipital cortices were preserved in their entirety, as were basal ganglia and thalami.

What neuroscience can and cannot tell us

We should not reverse-infer a psychological process from regional activation alone

- **Neuroimaging can only be correlated with behaviour**
- Brain regions do **not** map one-to-one onto psychological functions
- Amygdala and vmPFC are parts of **distributed, interacting systems**
 - Other areas activated during emotion processing
 - Same areas activated during non-emotion related activities
- The same behavioural failure can result from disruption at **different stages of processing**
- Lesion studies are strongest for showing that a neural system makes a **causal contribution** to behaviour

Reverse inference

Experiment	Previous fMRI Study	Current fMRI Study
Experimental Manipulation	Task: View! 	Task: Choose! You Receive € 50 KEEP € 20 Keep All Lose All GAMBLE You Receive € 50 LOSE € 30 GAMBLE

We are inferring the cognitive process:

1. Amygdala was activated during emotion processing in past research
2. When I tested participants, the amygdala was activated
3. Therefore, I am seeing emotion processing

This is not precise and potentially inaccurate: We do not have this much specificity in these measurements

Skin conductance response (SCR)

Measures: sympathetic sweat-gland activity **good index of autonomic arousal in response to a stimulus**

- Stimulus/event → sympathetic activation → sweat-gland activity → increase skin conductance
- Shows up even when participants don't self-report a change
- Sensitive to anticipatory responses
- Cheap, non-invasive, well-established
- BUT: Increasing skin conductance ≠ “emotion”
- A SCR can occur because something is: frightening, exciting, surprising, anticipated, novel, attention-grabbing, cognitively demanding, personally significant, painful, behaviourally relevant, etc. etc.
- Conditions/stimuli must be precisely matched: **CONFOUNDS**
- SCR provides evidence that the event altered autonomic responding
- Psychological meaning must come from the experimental contrast.

Brain areas & mechanisms: amygdala & VMF

- **Bechara et al. (1999): where does the failure occur?**
- Compared: bilateral amygdala damage ($n = 5$), bilateral VMF damage ($n = 5$), and healthy controls ($n = 13$).
- Test 1: Iowa Gambling Task + skin conductance (SCR)
 - **Can the emotional consequences of previous choices be used to guide later decisions?**
 - Participants **choose** a deck leading to a **monetary gain or loss**. Over repeated choices, they can learn which decks tend to produce bad long-term outcomes.
 - SCR measured **after the gain/loss** and **before the next choice**.
- Test 2: Learning task - Classical Conditioning
 - **Can they form a basic affective association between a predictive cue and an unpleasant outcome?**
 - A neutral visual cue is repeatedly followed by a loud unpleasant sound.
 - If the association is **learned**, the visual cue alone eventually produces an SCR.

Where in the process does the problem occur?

Amygdala damage: impaired acquisition and production of affective signals – significance not assigned to the event

VMF damage: affective responses are still generated, but are not used to guide choice

	Healthy controls	Amygdala damage	VMF damage
Learn advantageous decks?	✓	✗	✗
SCR after wins/losses?	✓	✗ *reduced	✓
Anticipatory SCR before choice?	✓	✗	✗
Conditioned SCR to visual cue?	✓	✗	✓

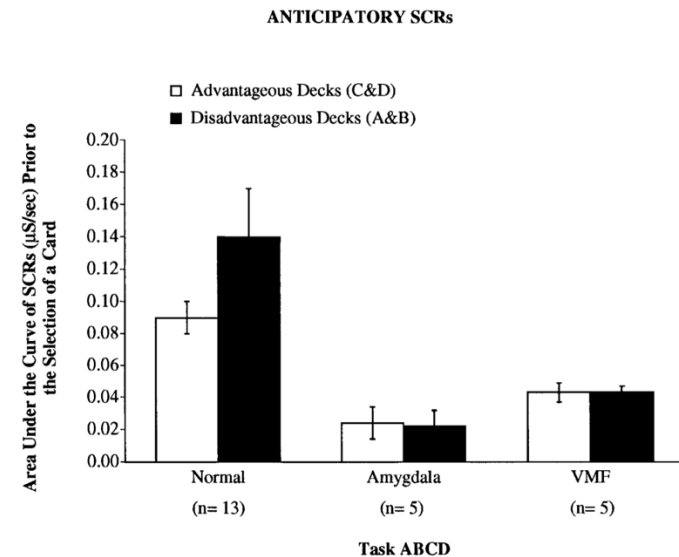


Figure 3. Means $\pm$ SEM of anticipatory SCRs (μ S/sec) generated by controls, amygdala, or VMF patients in association with the advantageous decks (C and D, white columns) versus the disadvantageous decks (A and B, black columns).

What does SCR show?

- Does autonomic arousal (measured by skin conductance) **cause** the judgement (this is risky) and lead to the avoidance behaviour?
 - This is the kind of role proposed by Damasio's somatic marker hypothesis.
- Does autonomic arousal occur alongside the risk judgement? And then one or both of these may lead to the avoidance behaviour?
 - This shows SCR predicts choice, but it doesn't imply that it causes choice

What is actually going wrong?

- **IOWA GAMBLING TASK: Choose the “good decks”**
 - **A/B:** larger immediate gains but large losses → bad long-term decks
 - **C/D:** smaller gains but smaller losses → advantageous long-term decks.
- Poor performance could reflect problems with:
 - attending to relevant information;
 - learning associations;
 - generating/evaluating options;
 - deliberation;
 - value assignment;
 - recognising that a decision is needed;
 - translating a choice into behaviour...
- Most of these processes are **not separately measured**.
- It is hard to conclude something more specific like: *somatic markers are required for evaluating alternatives*.

Enter the murky waters

- Emotion: poorly defined
 - Decision processes: poorly defined
 - Measurement tools: not perfect
 - Theories: still require development
-
- A lot of things are happening at the sameish time

Before* decision-making

*and also during ... and after ...

Emotion guides attention

Affect changes what information gets priority before and during decision-making.

- **Attention = selective processing:** we cannot process everything equally – something must be prioritised
- What receives attention is more likely to be **processed further and encoded into memory**
- Emotion can change **which information receives this priority**
- **Some key effects:**
 - Priority for high arousal stimuli
 - Priority for negative stimuli
 - Mood-congruent attentional biases

Anderson (2005): The Attentional Blink

- Participants viewed **rapid** serial visual presentation of words and asked to report the two **green words**
 - Matched words on length, frequency, neighbourhood frequency
 - Manipulated valence, arousal
- **At short T1–T2 lags, identifying T1 normally impairs detection of T2**
 - Effect is alleviated for **emotionally arousing words**
 - Emotional significance can influence what gets selected even when attentional resources are limited

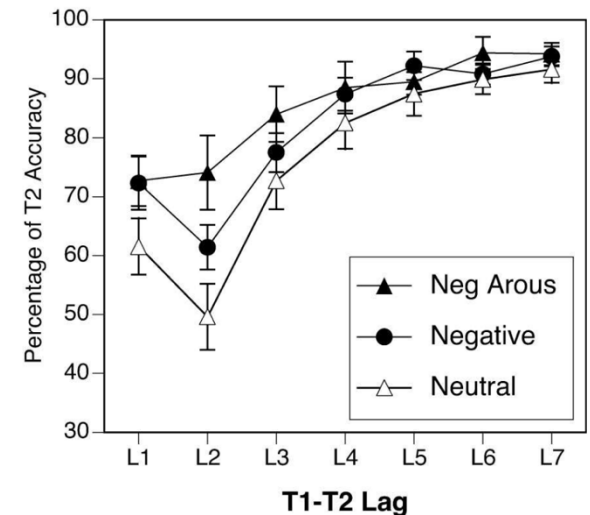
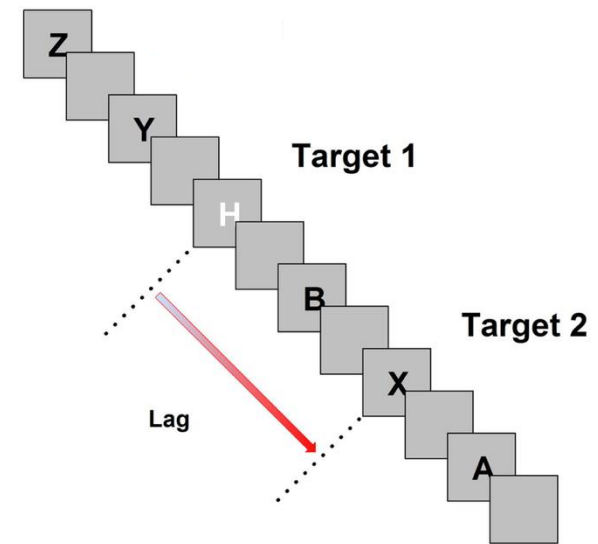


Figure 1. Mean percentage of accurate second target (T2) report given first target (T1) report (T1/T2) at each T1-T2 lag (L) in Experiment 1. Error bars represent standard errors of the means. Negative = negative and lower arousal T2 words; Neg arous = negative and highly arousing T2 words; Neutral = neutrally valenced and low-arousal T2 words.

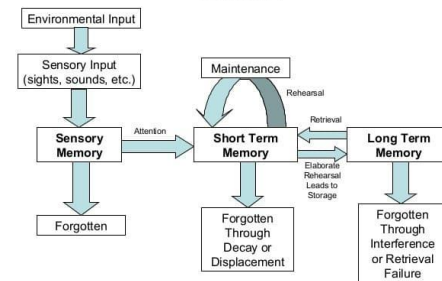
Emotion guides memory

- **Memory = processes involved in encoding, retaining and later retrieving information**

Encoding = information is represented in a form that can enter memory

Retrieval = stored information becomes available again later

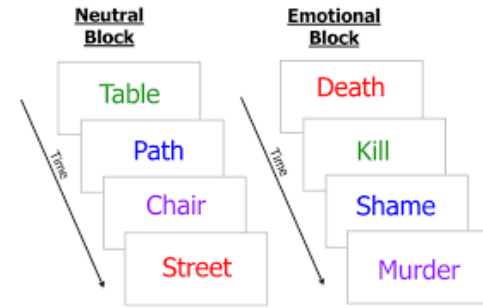
Multi Store Model - Atkinson & Shiffrin



- Emotional material often has a **memory advantage**
 - Emotional arousal enhances long-term memory: participants remembered an emotionally arousing story better than a matched neutral version
 - Giving participants **propranolol** reduced the emotional-memory advantage → evidence that the effect depends partly on **adrenergic arousal** (Cahill et al., 1994).
- But the effect is **selective**, not global:
 - Emotional events can produce a **memory trade-off**: strong memory for emotional material but poorer memory for surrounding information (Christianson & Loftus, 1991; Kensinger et al., 2007).

Taboo Stroop Task

- **Stroop task:** Name **ink colour**, ignore word meaning
- Taboo words → slower colour naming
 - Meaning interfered even though it was task-irrelevant
- **Lexical decision:** speeded decision task - is it a real or pseudoword?
 - No equivalent general slowing in lexical decision
- Task dependent effects: emotional significance changes the allocation of processing, rather than producing a generic lexical-processing cost
- **Later recall:** Taboo words were **better remembered**
 - But they impaired immediate memory for **surrounding neutral information**
- See: MacKay et al. (2004)

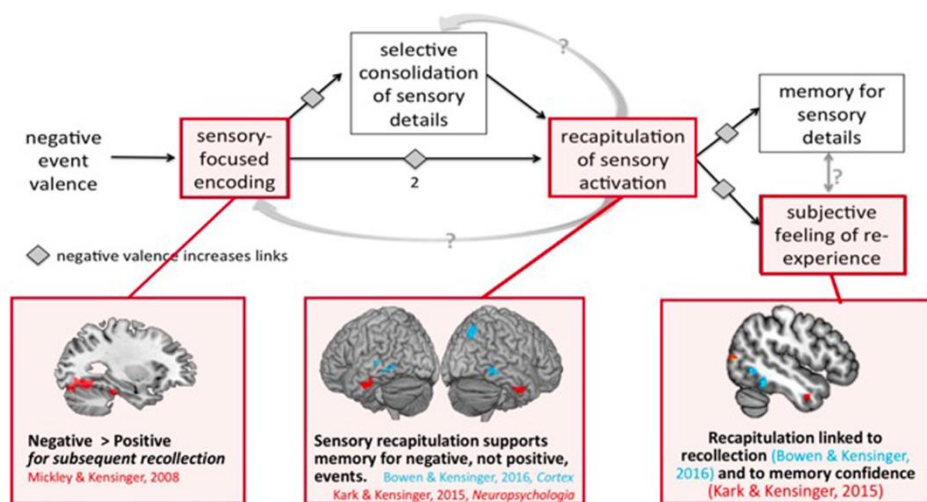


Valence-specific memory model

- Most emotional-memory theories focus heavily on **arousal**.
- Bowen et al. (2018) argue that **valence itself matters**: even when negative and positive events are matched for arousal, negative events can show different behavioural and neural memory profiles.

Negative Emotional Valence Enhances Recapitulation (“NEVER Forget”)

Bowen, Kark, & Kensinger, *Psychonomic Bulletin & Review*, in press



Psychon Bull Rev (2018) 25:870–891

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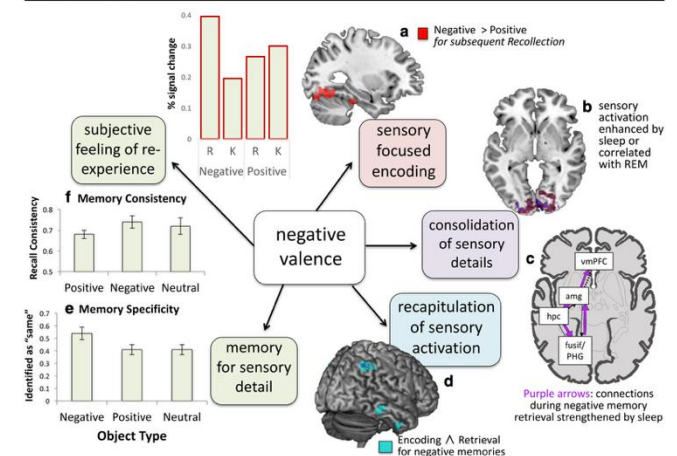


Fig. 1 The neural processes and behavioral outcomes associated with memory for negative events. Negative valence has been shown to enhance the encoding, storage, and recapitulation of sensory details and to lead to enhancements in memory for sensory detail and in the subjective feeling of re-experiencing an event. Examples of these effects,

from the authors' prior research, are depicted. Data from (clockwise): (a) Mickley and Kensinger, 2008; (b) Benmion et al., 2017; (c) Payne and Kensinger, 2011; (d) Kark and Kensinger 2015a; (e) Kensinger, Giamff Eaton, et al. 2007b; (f) Holland & Kensinger, 2012

Decision-making

Affect as information

- Instead of trying to calculate the objective value of an outcome, people may just ask themselves:
 “how do I feel about this?”
- Feelings are treated as **evidence / information**
- Their effect depends on whether they **seem** relevant, representative, and attributable to the target
 - Emotions (or moods) not related to the task at hand *spill over* and affect decision-making
 - Experienced at the moment of choice but stem from **dispositional or situational sources** that are orthogonal to the task at hand
- See: Clore et al. (2001)
- For reviews: Han et al., 2007; Keltner & Lerner 2010; Loewenstein & Lerner 2003



Incidental emotions



- Broadly, people in **good moods make optimistic judgments**, and people in bad moods would make pessimistic judgments
 - positive mood → positive judgement
- **Johnson & Tversky (1983)**: people who read negative stories offered pessimistic estimates of fatalities in a subsequent unrelated task
- **Schwarz & Clore (1983)**: ambient weather influenced people's self-reported life satisfaction
 - Until they were told to assess where their feelings came from!

* Later replications found much smaller effects

How satisfied are you with your life?

I feel bad right now

Why do I feel like this?

Source misattributed:
**Must be because of my
life**



Use feeling to decide



Mood impacts
judgement:

I hate my life!

Source identified:
**because the weather is
shit**



Discount feeling as
irrelevant



Mood effect on
judgement reduced:

My life is fine

Affect Heuristic: Risk Judgements

How do I feel about: NUCLEAR POWER?

- *Dangerous? Frightening? Catastrophic?*
- Leads to an **overall affective impression: BAD**
- Helps answer: How risky is it? How beneficial is it?
- When judging a complex object or option, people can use their **overall affective reaction** as a shortcut
 - “Dipping into the affect pool”
but what does that mean???
- Some risks evoke disproportionate fear because they seem **catastrophic, uncontrollable and fatal** (Slovic, 1987).
- See: Finucane et al., 2000; Slovic et al., 2007

Affect Heuristic

- **Finucane et al. (2000):** affect becomes a shortcut for complex judgement
 - More pronounced under time pressure
 - Implications: **our cognitive resources are often limited**
- **When we feel good about something we decide it has high perceived benefit and low perceived risk (and vice versa)**
- Why should perceived risk and benefit be negatively correlated? In the real world, they often aren't

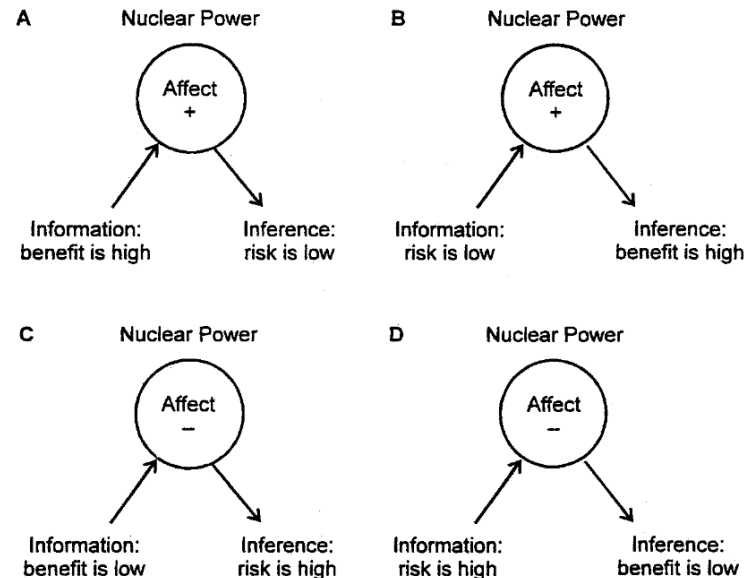


Exhibit 4. Model showing how information about benefit (A) or information about risk (B) could increase the global affective evaluation of nuclear power and lead to inferences about risk and benefit that are affectively congruent with the information input. Similarly, information could decrease the global affective evaluation of nuclear power as in C and D, resulting in inferences that are opposite those in A and B

Risk-as-feelings

- Risk has two components:
 - what we **calculate** about probability and severity
 - how much danger we **feel**.
- Feelings do not perfectly track calculated risk
 - vivid, immediate threats can feel very dangerous even when probability is low.
- **When calculated risk and felt risk conflict, decisions may follow the emotional response**
- **Competing goals & fast decisions: tricky**

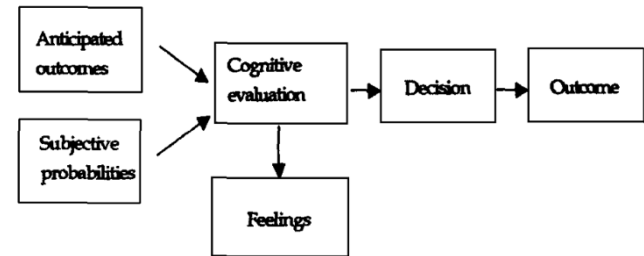


Figure 1. Consequentialist perspective.

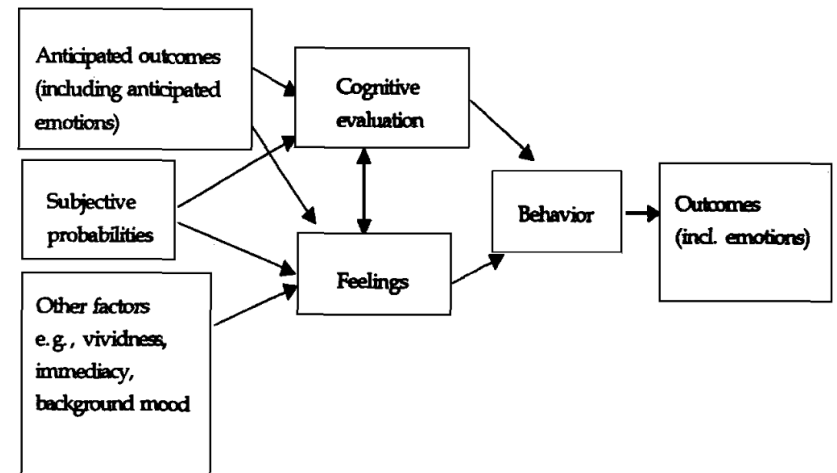


Figure 3. Risk-as-feelings perspective.

(Loewenstein et al. (2001)

Beyond valence: Fear, anger & happiness

- **Lerner & Keltner (2001):**
 - EXP1 ($n = 75$) and EXP2 ($n = 601$): dispositional anger/fear
 - EXP3 ($n = 118$) – dispositional anger/fear/happiness
 - **EXP4** ($n = 60$) – induced fear or anger by writing vividly about a personal event
 - Subsequently estimated risks and rated appraisals including certainty and control
- FEAR → uncertainty + low personal/situational control
- Fear → “bad outcomes are more likely”
- ANGER → certainty + high personal control
- Anger → “bad outcomes are less likely” greater optimism

Fear, anger & happiness

- Lerner & Keltner (2001) found that estimates of angry people more closely resembled those of happy people than those of fearful people.
- Replication ($n = 780$) found support for:
 - risk optimism in general,
 - risk optimism for positive events
 - risk optimism for ambiguous events,
 - but not for risk preference and risk optimism for negative events (Lu et al., 2025)

Moral decision-making

- Does affect change whether people endorse **instrumental interpersonal harm**?
- A purely outcome-based calculation might say: **1 death < 5 deaths → sacrifice one to save five**
 - people often reject that action when the better aggregate outcome requires **directly harming someone**
- The final outcome is therefore not sufficient to predict the judgement
- Moral decisions integrate information about **harm, action, intention, causation and consequences**
- Affective aversion to interpersonal harm can be one input into that integration

Koenigs et al. (2007):

- **Damage to the prefrontal cortex increases “utilitarian” moral judgements**
- **Participants with bilateral ventromedial prefrontal cortex (VMPC) damage ($n = 6$), with 12 brain-damaged controls and 12 healthy controls**
- **Participants judged 50 hypothetical scenarios: 18 non-moral, 11 impersonal moral, 21 personal moral,**
- **For personal moral dilemmas, VMPC patients were significantly more likely to endorse the sacrificial action compared to:**
 - healthy controls: **OR = 2.81**
 - brain-damaged controls: **OR = 3.30.**
- **This was even more pronounced for high-conflict personal dilemmas:**
 - VMPC damage vs healthy controls: **OR = 4.70**
 - VMPC damage vs brain-damaged controls: **OR = 5.38.**

New issue: Moral theories

Moral Objectivity: Valence Asymmetry

Goodwin and Darley (2012): Is there a correct answer to this moral question, or is it a matter of opinion?

- Beliefs about the **wrongness of negative acts** were judged more objective than beliefs about the **goodness of positive acts**
 - Not a virtue of strength of endorsement

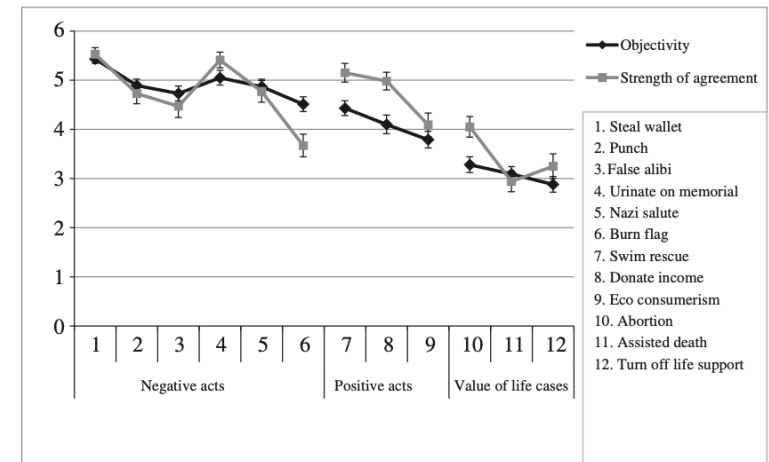


Fig. 1. The mean objectivity and strength of agreement ratings for each of the 12 different moral items in Study 1.

Objectivity pattern:

- Facts, Negative moral acts, positive moral acts, contested issues, social norms, taste

Consensus & Objectivity

- If I think a moral belief is **objectively true**, then there must be a correct answer – this is not a preference
- Study 1: greater perceived objectivity predicted greater discomfort with a person who disagreed
 - Greater objectivity also predicted judging the dissenter as **more immoral**
 - And being more certain of your own choice
- These relationships remained after accounting for strength of agreement
- Study 2 ($n = 70$) experimentally manipulated perceived consensus:
 - higher apparent consensus produced **higher perceived objectivity**
- Suggesting a link between: evaluation, consensus and social judgment of dissenters

Goodwin & Darley: Issues

- Goodwin & Darley used different response scales
 - It's possible *wrong/right* expresses a **rule or obligation**, while *good/bad* expresses evaluation
 - CONFOUND
- Goodwin & Darley used different scenarios
 - Cost of action? Social rules? CONFOUND
- **The replication crisis strikes again:**
 - We haven't been able to replicate this effect with a more controlled set of dilemmas

“Moral judgement” is not one dependent variable

- We can ask whether an action is :
 - **Wrong?**
 - **Blameworthy?**
 - **Permissible?**
 - Whether someone **should** perform it
 - How strongly the agent should be **punished**
- These questions recruit overlapping information, but they are not interchangeable measures of one latent response

The wording of the judgement question is part of the experimental manipulation

- **Wrongness $\neq$ blame $\neq$ punishment** → the dependent variable changes what psychological process you are measuring
- See: Malle (2021) for a review

Consider

- Mechanistic evidence requires measures or manipulations that discriminate plausible pathways
- Language can manipulate **affect**
- But also: **comprehension, fluency, imagery, attention, cultural frame, etc.**
- These pathways can operate simultaneously
- Therefore: a behavioural difference between two language conditions is not itself evidence that affect caused it
- This is especially important in cross-linguistics and bilingual research

Language impacts judgement

- **Fausey & Boroditsky (2010):** participants saw intentional and accidental events described with different linguistic frames
 - Agentive wording makes the causal agent explicit: “**He broke the vase**”
 - Non-agentive wording backgrounds the agent: “**The vase broke**”
- For accidental events, linguistic framing altered later memory for **who caused the event**
- So language can alter **upstream representations** that later judgement relies on

Decisions about others

- **Dunn & Schweitzer (2005):** 5 experiments
- Induced **incidental emotions** before participants made trust judgements about unrelated people.
- **Happiness and gratitude → more trust**
- **Anger → less trust**
 - Effects were stronger for emotions involving **other-person control** and weaker when the trustee was already familiar.
 - Weaker when they knew the person they evaluated
- How I feel can become information about whether another person is trustworthy, even when they did not cause the feeling.
- This is affect-as-information again, but now the target of the judgement is another person

Emotions: for better *and* worse

- Emotions, affect and mood affect decision-making in many ways
- They can improve decision-making
 - And are likely necessary for some decision-making
 - **Useful:** anticipated consequences, risk, regret, learned value...
- They can also make it worse
 - **Not so useful:** loss aversion, incidental emotions, excessive commitment, forecasting errors, impulsive choices...
- Emotions (like cognitive strategies) are not rational or irrational per se: How (un)reasonable their influence is
depends on their fit with the environment.

Emotion and decision-making

- “Does affect matter?” is usually the wrong experimental question.
 - The question is **where**, **when**, and **by what mechanism** affect changes interpretation or behaviour.
- How affect is operationalised also matters
 - Affect? Only valence? Mood? Specific emotions?
- “Which decisions are affected?” is a good starting point, but also not specific enough
 - Decisions are complex tasks with many stages

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